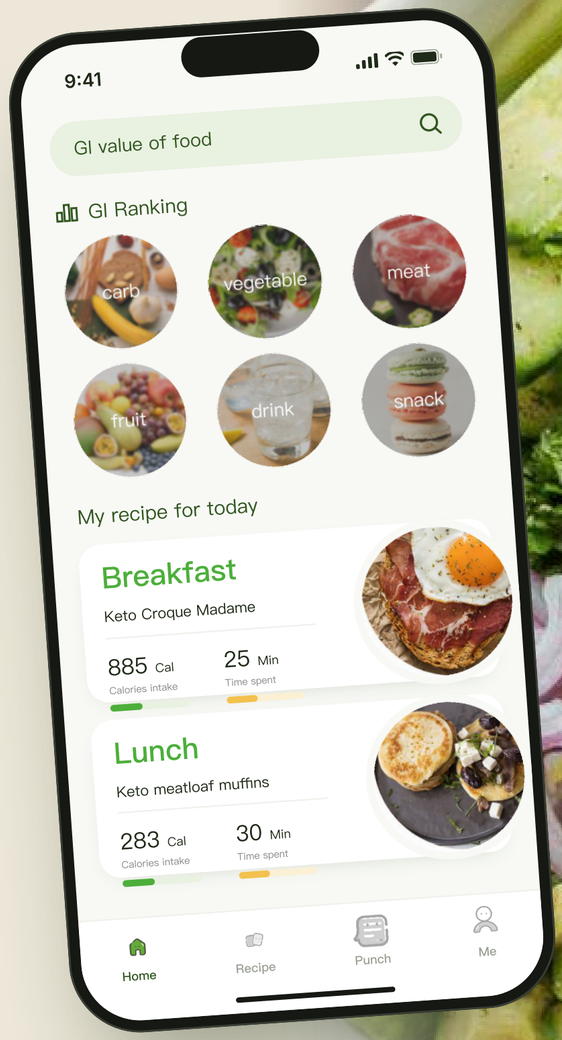
MOBILE APP · NUTRITION · RESEARCH-LED DESIGN

# Ketogenic *diet*

An app for *ketogenic dieters* — it answers the daily question of *what to eat*, and makes every day's nutrient intake *visible at a glance.*




INDEPENDENT AI CREATOR & DESIGNER

**Jasmine / Ming Gao**

SCOPE

**Research · IA · UI · Branding**

THE BRIEF

# Eating keto means reading *every label.* This app does the reading for you.

The ketogenic diet removes almost all carbohydrates, so dieters must monitor every meal. Existing apps felt complex, generic and tedious to log. This project turns research on real dieters into three simple jobs: know what's low-GI, get a plan you can follow, and see your progress.

**~100**
forum posts analysed

**18**
days of self-experimentation

**11**
interviews — 1 expert, 10 dieters

**16**
high-fidelity screens

CONTENTS


01 Background — What keto is & why it matters

02 Research — Forums · self-test · interviews

03 Insights — Journey map · pain → feature

04 Structure — Information architecture · low-fi

05 Visual Identity — Logo · colour · type · icons

06 Final Design — Home · Recipe · Punch · Me




**Jasmine / Ming Gao**
INDEPENDENT AI CREATOR & DESIGNER

# 01

# High fat, moderate protein, *almost no carbs.*

The ketogenic diet — commonly known as “keto” — supplies large amounts of fat, moderate amounts of protein and minimal carbohydrate, shifting the body from burning glucose to burning fat.

PLATE PROPORTION




NET CARBS PER SERVING · FEWER → MORE




**0g**
Meat · fish · natural fats

**1g**
Eggs · cheese

**1–5g**
Vegetables grown above ground

*Research and data from my EPQ paper “An Evaluation of the Ketogenic Dieting”.*

THE IMPORTANCE

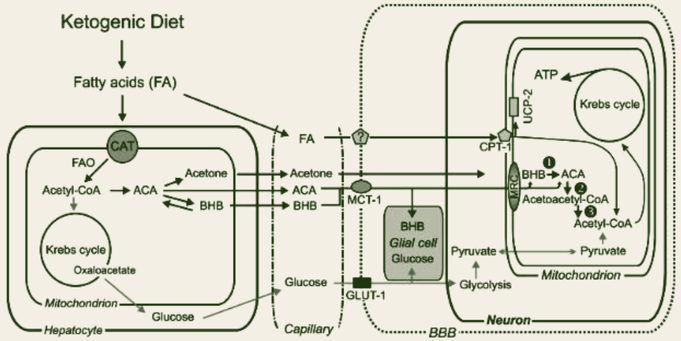
# Keto has gained significant attention for its potential to address *obesity, diabetes* and *neurological disorders.*

BENEFITS & RISKS

# A powerful diet — *with real trade-offs.*

## The metabolic pathway

Masino & Rho, 2012




+ BENEFITS

- Weight reduction
- Lower risk of cardiovascular disease
- Aids blood-sugar regulation for type 2 diabetes

– RISKS

- Nutrient deficiencies
- Disrupted gut flora
- Possible adverse cardiovascular effects
- Hard to sustain long-term

*Research and data from my EPQ paper "An Evaluation of the Ketogenic Dieting".*

# Four rules for eating keto *well.*

Cutting carbs is only half of it. A healthy keto plate still needs micronutrients, fibre, good fats and quality protein — exactly the guidance a dieter needs on hand every day.




01

## Vitamins & minerals

Include nutrient-dense, low-carb vegetables such as spinach, cauliflower and bell peppers.



02

## Fibre & low-starch veg

Spinach, kale and zucchini provide fibre and essential micronutrients.




03

## Healthy fats & omega-3

Avocados, olive oil, nuts and high-fat fish support heart health.




04

## Protein quality

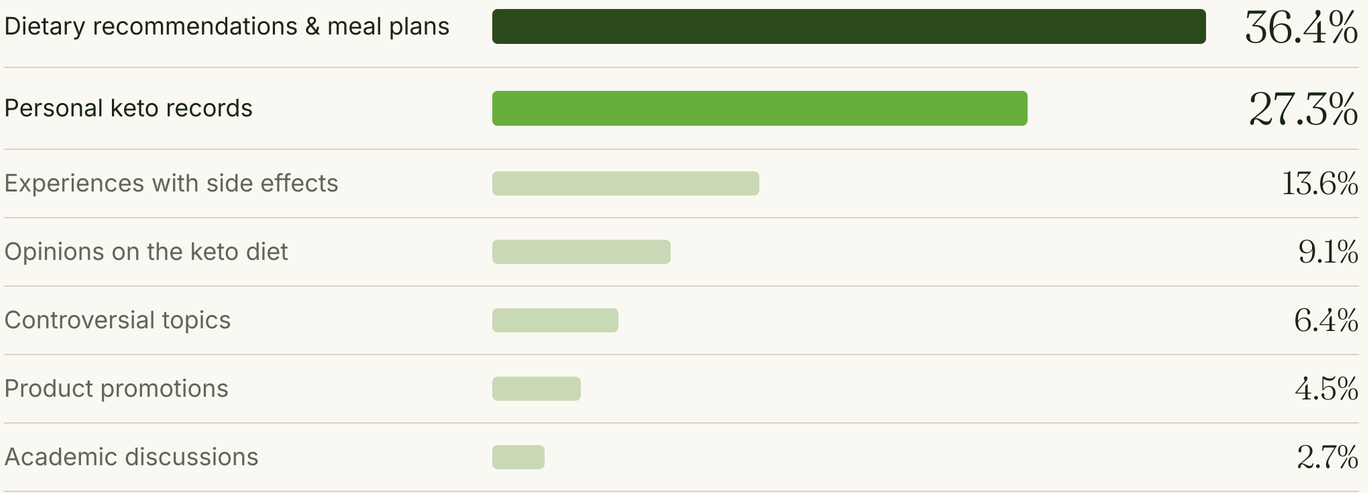
Adequate protein prevents muscle loss during weight loss on keto.

*Research and data from my EPQ paper "An Evaluation of the Ketogenic Dieting".*

# 02

# What are keto dieters *talking about?*

SHARE OF POSTS BY THEME




AIM

Investigate the nature of discussions around the ketogenic diet on social media forums.

SAMPLE

~100 newly published posts, coded into themes to find dominant trends in the keto community.

FINDING 01

Most discussion is about *recipes and meal planning.*

FINDING 02

Forums act as a place to *record and share* progress.

FINDING 03

“Lack of energy, insomnia, hair loss, low blood sugar...”

*Research and data from my EPQ paper “An Evaluation of the Ketogenic Dieting”.*

METHOD · DIARY STUDY ON MYSELF

# I went keto for two weeks *to feel it first-hand.*

2 preparation days, 14 keto days and 2 recovery days. Every day I logged fasting and post-meal blood glucose, sleep, exercise and how I felt physically and mentally.

BLOOD SUGAR

## Stable

Readings stayed consistent through the keto phase.

WEIGHT

## −2.6 kg

Lost over the 14-day experiment.

HOW I FELT

## *Keto flu*

Hunger, headaches, low energy and constipation early on.

TABLE 1 · DIARY DATA

Table 1 Diary data

| Day | Diet | Fasting Blood Glucose (mmol/L) | Postprandial Blood Glucose (mmol/L) | Sleep | Exercise | Physical and mental changes |
|---|---|---|---|---|---|---|
| Prep: 2 days | Normal | 4.9 | 6.0 | good | swimming | stable |
| Keto 1 | Keto | 4.9 | 5.2 | good | - | hungry |
| Keto 2 | Keto | 4.6 | 5.0 | good | - | - |
| Keto 3 | Keto | 4.7 | 5.1 | good | walk | Mild constipation |
| Keto 4 | Keto | 4.7 | 5.0 | worse | - | hungry |
| Keto 5 | Keto | 4.8 | 5.1 | moderate | swimming | High strung |
| Keto 6 | Keto | 4.7 | 5.1 | good | - | Mild headache |
| Keto 7 | Keto | 4.6 | 5.0 | poor | - | Low energy |
| Keto 8 | Keto | 4.6 | 4.9 | good | walk | hungy |
| Keto 9 | Keto | 4.5 | 4.8 | moderate | - | - |
| Keto 10 | Keto | 4.6 | 4.9 | good | walk | constipation |
| Keto 11 | Keto | 4.6 | 5.0 | poor | - | stable |
| Keto 12 | Keto | 4.5 | 4.9 | moderate | - | More focused |
| Keto 13 | Keto | 4.7 | 5.0 | good | swimming | Clear focus |
| Keto 14 | Keto | 4.6 | 4.9 | good | - | - |
| Post Keto: 2 days | Normal | 4.6 | 6.0 | poor | walk | - |

*Research and data from my EPQ paper "An Evaluation of the Ketogenic Dieting".*

1 EXPERT · 10 DIETERS

# One diet, *very different* dieters.

DIETERS | 10 participants | 5 women · 5 men | Ages 18–30 · 31–49 · 50+ | Social-media keto groups

Q1 · WHY DID YOU START?

## Age shapes motivation

Older participants started for health concerns such as pre-diabetes or weight management. Younger participants were driven by lifestyle goals like weight loss.

Q2 · HOW DID YOU PREPARE?

## Gender shapes preparation

Women planned meals and consulted a professional first. Men started with minimal preparation, relying on online communities.

EXPERT · NUTRITIONIST, 8 YEARS

“Keto is an almost total elimination of carbohydrates — it requires *meticulous monitoring* of food intake.”

**Not just low-carb:** close to no carb, so food labels must be checked again and again.

**Who it suits:** epilepsy, PCOS, high blood lipids, blood-sugar control, overweight.

**Advice:** check suitability first, then reduce carbs gradually — not to zero at once.

SUMMARY

Motivation, preparation and staying power all differ — so the app has to *adapt to the individual*, not the average.

**6–12 mo**

Typical time most participants stayed on keto.

***Prepared → better***

Older dieters benefited from thorough preparation; younger ones relied on willpower.

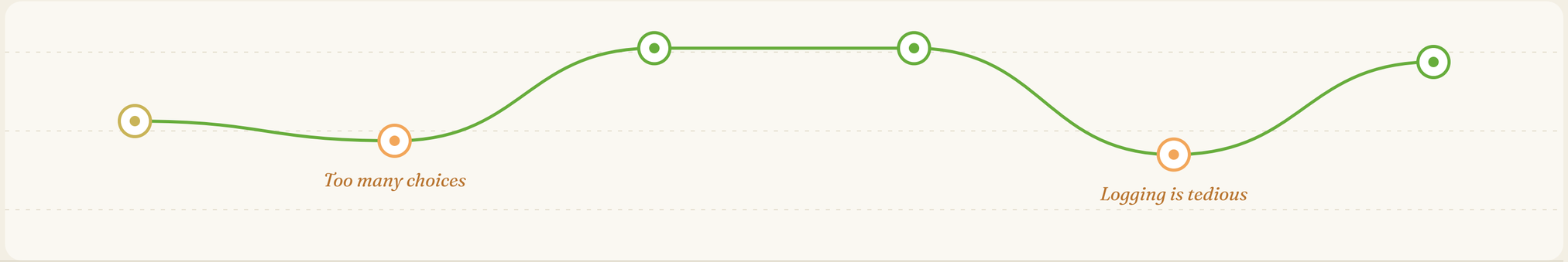
***No one-size plan***

Individual differences must shape goals, menus and guidance.

*Research and data from my EPQ paper “An Evaluation of the Ketogenic Dieting”.*

# Where today's apps *lose people.*

Mapping a typical dieter's journey through existing diet apps shows two emotional dips: choosing an app, and the day-to-day effort of logging food.




| STAGE | Need | Search | Find | Use · Learn | Use · Log | After use |
|---|---|---|---|---|---|---|
| EXPERIENCE | I often lack energy at work<br>Worried about the health effects of excess weight | Weight-loss app recipes feel complex<br>Too many similar apps to choose from | Search “keto diet”, “healthy diet”, “weight loss”<br>Clear function description and guide | Clear, concise guide for beginners<br>Clear recipe categories and search | Recipes ignore personal taste and nutrition; steps complicated<br>Logging food is tedious | Frequent updates, but new features aren't practical<br>Nothing new in the recipes |
| EXPECTATION | *Find an app for weight loss and energy* | *See positive reviews and high ratings* | *Easy to use* | *Recipes based on preferences and health goals* | *Clear nutrient analysis and progress tracking* | *Regular, genuinely useful new features* |

# Three pains, *three answers.*

| PAIN POINT | REQUIREMENT | FEATURE |
|---|---|---|
| The app interface is too complex | → Clear, direct functions | *Simplified home* |
| “What should I eat?” every day | → Targeted recipe recommendations | *Custom recipes* |
| Food GI information is rough | → Keep sugar intake to a minimum | *Punch-in & visualise* |

BRAINSTORM · FEATURE MAP

## Keto *Diet*

Every idea from the brainstorm, clustered into four pillars that became the app's structure.

### *Home*

- Home reminder
- GI ranking
- Nourishment

### *Recipe*

- Food classification
- Search recipes
- Recommended · Custom

### *Plan*

- Lose / maintain weight
- Muscle-building
- Seven-day diet · Add plan

### *Visualisation*

- Clock-in · Schedule
- Daily completion
- Weight-change curve

*Research and data from my EPQ paper “An Evaluation of the Ketogenic Dieting”.*

# 04

# Four tabs. *Nothing extra.*

After onboarding collects body data and goals, everything lives in four tabs — each one maps directly to a research-driven feature.

## Enter
ONBOARD

- **Log in**
- **Personal information**
  - Weight & height
  - Gender & age
  - Goal

## Home
TAB 01

- **Food GI ranking**
- **My daily recipe**
  - Breakfast
  - Lunch
  - Dinner

## Recipe
TAB 02

- **Recommended menu**
- **My recipes**
  - Create a menu
  - DIY a recipe

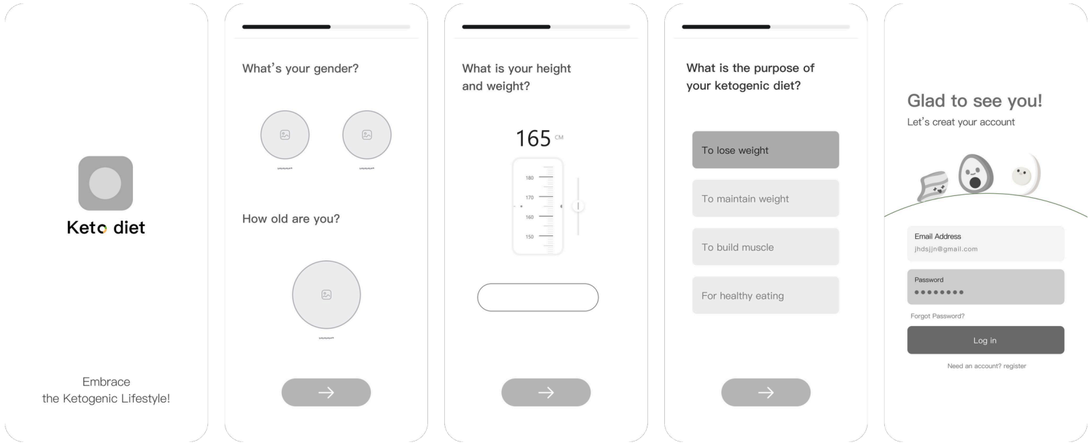
## Punch
TAB 03

- **Daily task**
  - Three meals a day
  - Calorie intake
  - Carb intake
- **Punching**

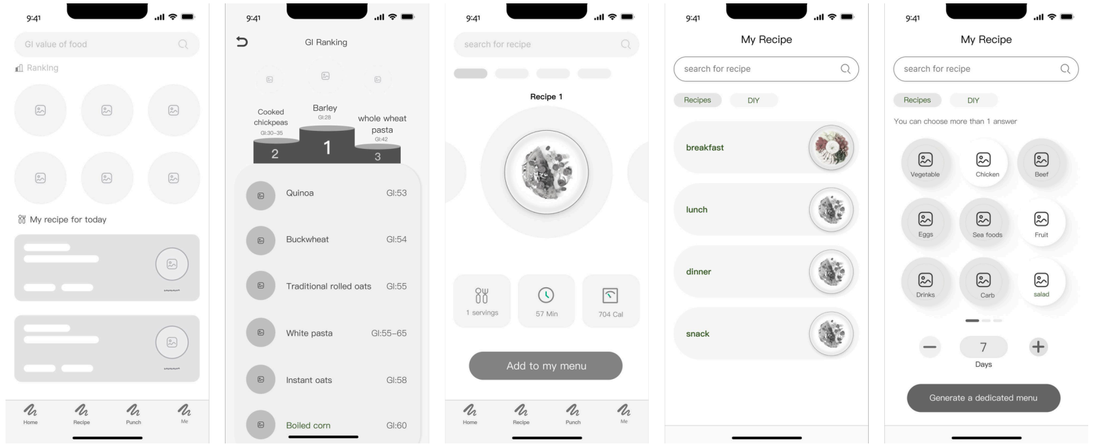
## Me
TAB 04

- **Personal data**
  - Data curve
  - Settings

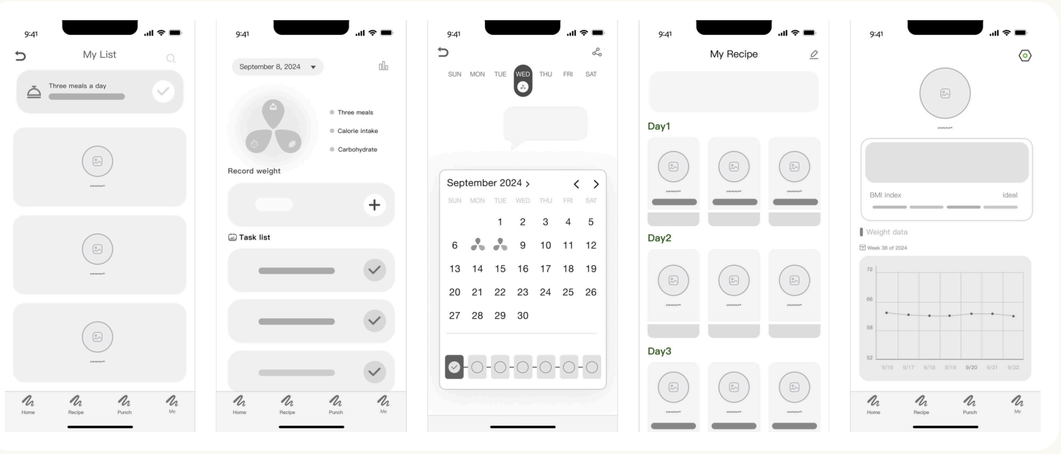
A · ONBOARDING




B · HOME & RECIPE




C · PUNCH & ME




LOW FIDELITY

# Wireframes first, *flows before pixels.*

Fifteen greyscale screens tested the full path — onboarding, GI lookup, recipe building and daily punch-in — before any colour or imagery was applied.

LOGO



# Keto diet

A surprised avocado — keto's signature fat — and a nutrient-ring "o" that echoes the data at the heart of the app.

COLOUR

**Forest**
#325721

**Leaf**
#67AD3C

**Mint**
#F0F7EB

Calorie · Carb accents

TYPE · PINGFANG SC

Aa Heavy 20px

Aa Medium 16px

Aa Regular 12px

ICONS & MASCOT



Home Recipe Punch Me Avo

# 06

# Know your food *before you eat it.*

**1 GI search, front and centre**

The fastest way to check a food's glycaemic index — the question keto dieters ask most.

**2 Browse by category**

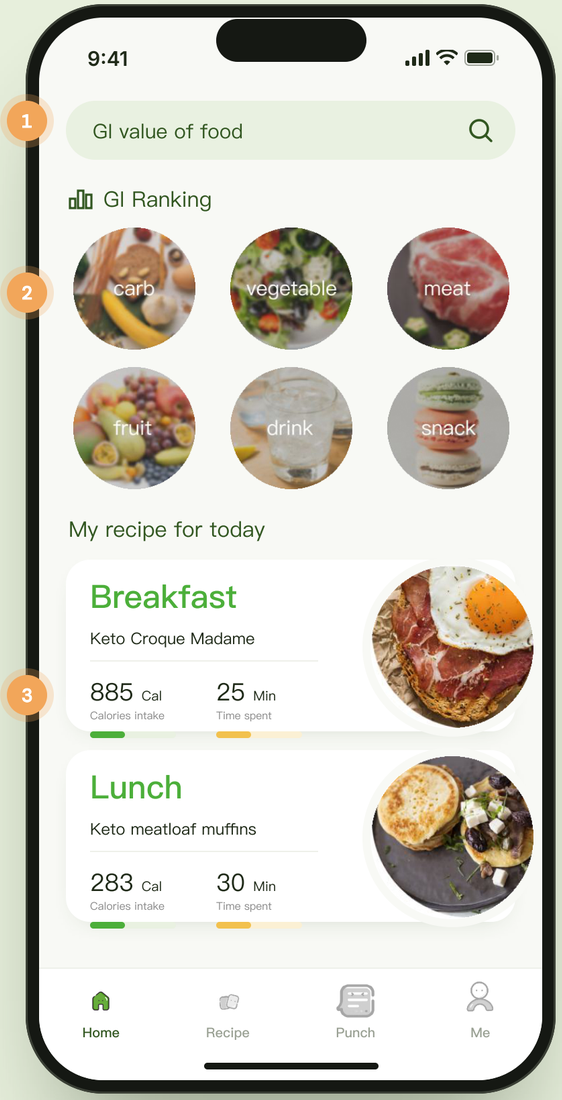
Six photo categories lead to ranked lists, so low-GI choices surface first.

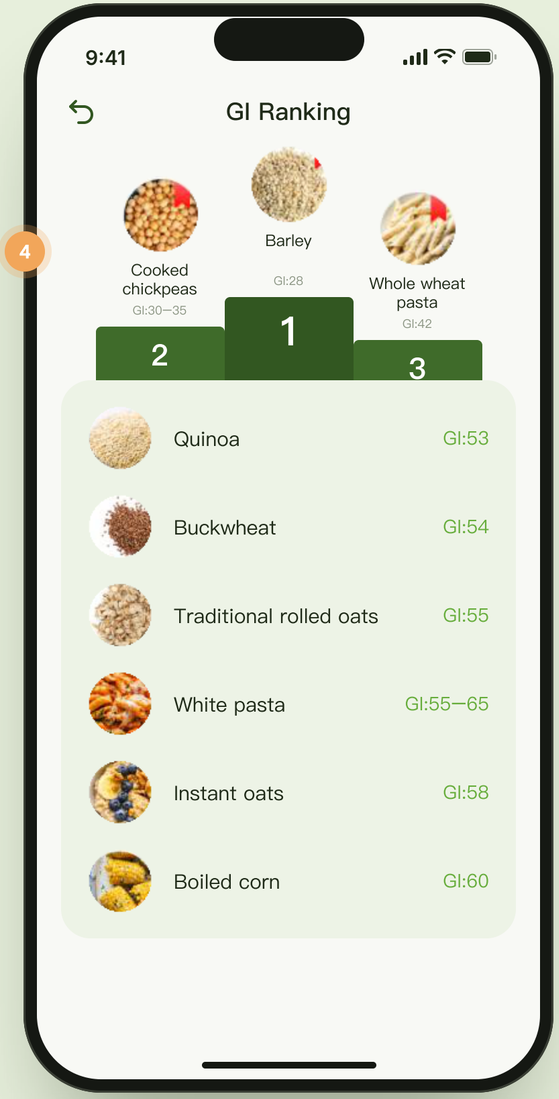
**3 Today's menu as a reminder**

The chosen plan appears on Home with calories and cooking time.

**4 A podium for the lowest GI**

The top three are celebrated; the rest are listed with exact GI values.



# Build a plan *in three taps.*

**1 DIY by ingredient**

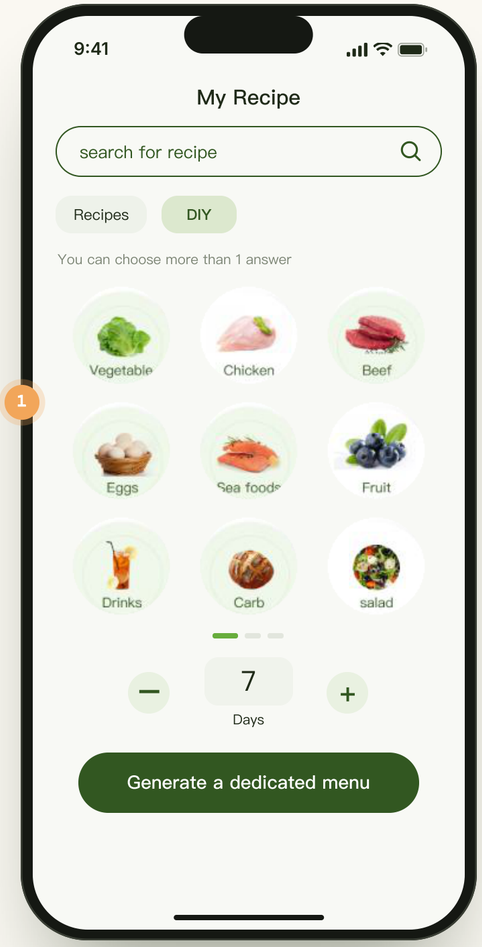
Pick the foods you like, set how many days.

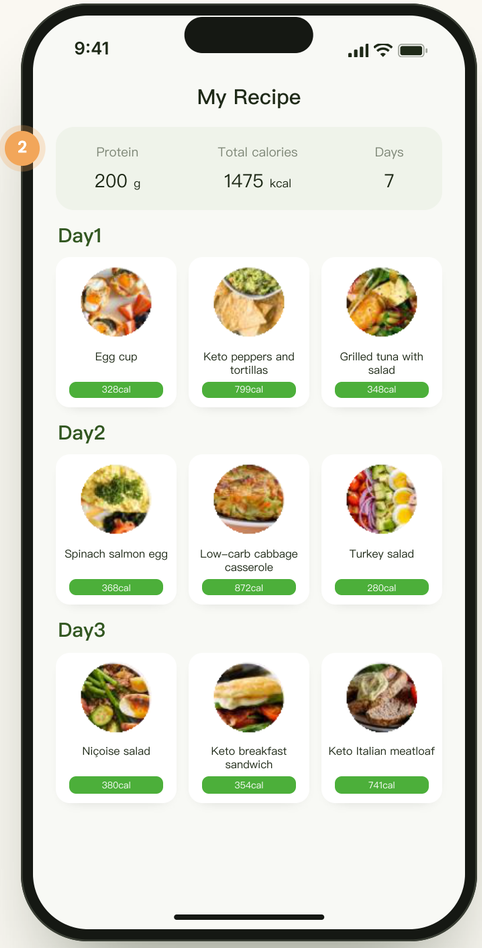
**2 A dedicated menu**

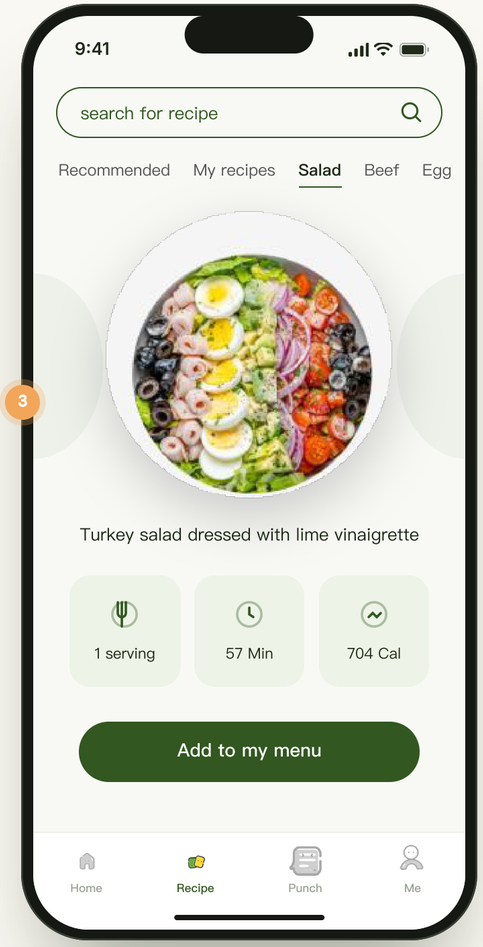
Day-by-day meals with protein and calorie totals.

**3 Or pick a recipe**

Servings, time and calories — then add to menu.



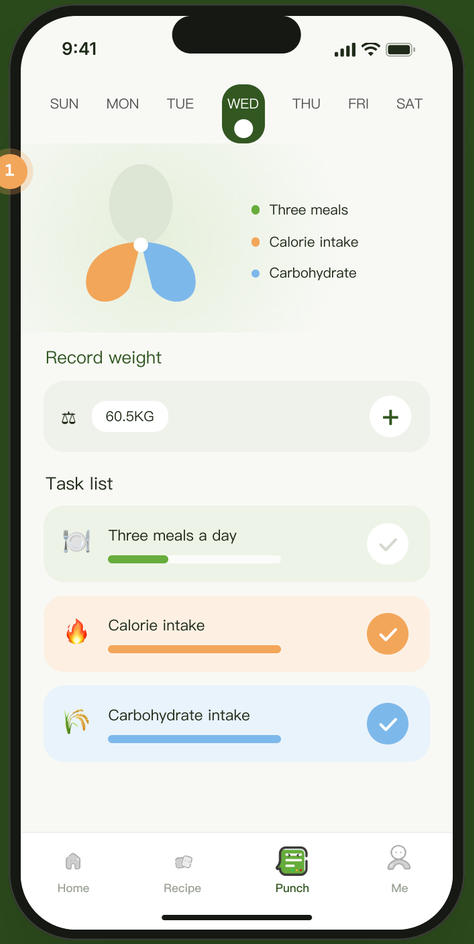



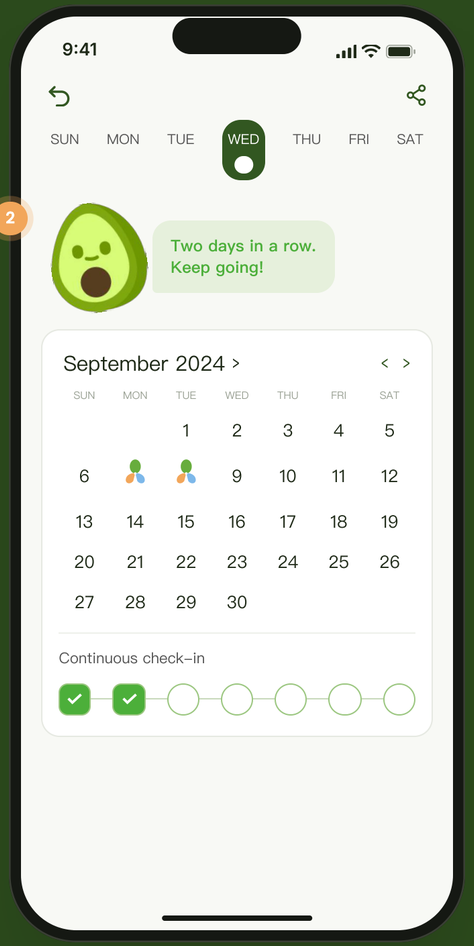
# Progress you can *see bloom.*

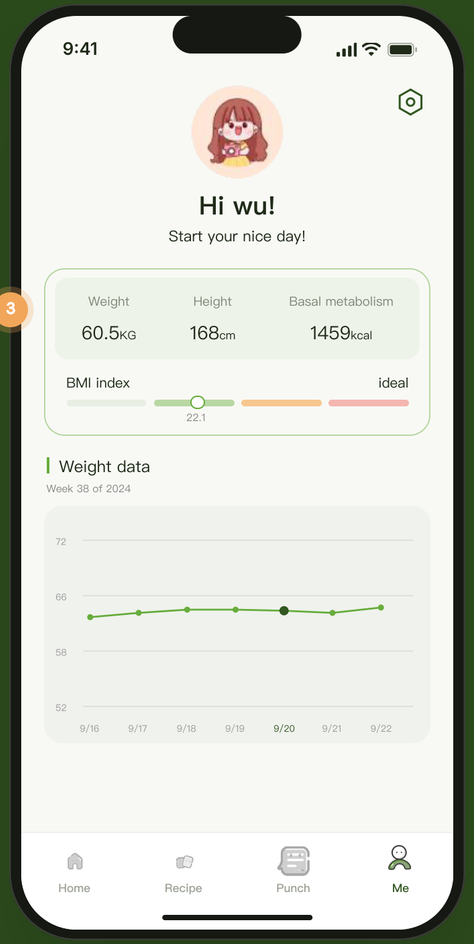
**1 The three-colour flower**
Meals, calories and carbs fill in as petals through the day.

**2 Streaks with Avo**
Completed days bloom on the calendar; Avo cheers you on.

**3 Your body, in data**
Weight, BMI, metabolism and the weekly weight curve.



16 SCREENS · END-TO-END FLOW

# Thank *you.*


Jasmine / Ming Gao

INDEPENDENT AI CREATOR & DESIGNER